

$\mathcal{K}_0^{#1} \uparrow$	$\mathcal{K}_0^{#1} \alpha\beta_X$				
$\frac{1}{2} (-Y_1 k^2 - 3 \binom{(2)}{K_3})$	0				
$\mathcal{K}_0^{#1} \uparrow^{\alpha\beta_X}$	0	$\mathcal{K}_{1^+}^{#1} \alpha\beta$	$\mathcal{K}_{1^+}^{#2} \alpha\beta$	$\mathcal{K}_{1^-}^{#1} \alpha$	$\mathcal{K}_{1^-}^{#2} \alpha$
	$\mathcal{K}_{1^+}^{#1} \uparrow^{\alpha\beta}$	$\frac{1}{2} (-Y_3 k^2 + 4 \binom{(2)}{K_1})$	$\frac{-Y_3 k^2 + 4 \binom{(2)}{K_1}}{2 \sqrt{2}}$	0	0
	$\mathcal{K}_{1^+}^{#2} \uparrow^{\alpha\beta}$	$\frac{-Y_3 k^2 + 4 \binom{(2)}{K_1}}{2 \sqrt{2}}$	$\frac{1}{4} (-Y_3 k^2 + 4 \binom{(2)}{K_1})$	0	0
	$\mathcal{K}_{1^-}^{#1} \uparrow^\alpha$	0	0	$\frac{1}{2} (Y_4 k^2 - 2 \binom{(2)}{K_3})$	$\frac{Y_8 k^2 + 2 \binom{(2)}{K_1}}{2 \sqrt{2}}$
	$\mathcal{K}_{1^-}^{#2} \uparrow^\alpha$	0	0	$\frac{Y_8 k^2 + 2 \binom{(2)}{K_3}}{2 \sqrt{2}}$	$\frac{1}{4} (-Y_6 k^2 - 2 \binom{(2)}{K_3})$
				$\mathcal{K}_{2^+}^{#1} \alpha\beta$	$\mathcal{K}_{2^+}^{#1} \alpha\beta_X$
	$\mathcal{K}_{2^+}^{#1} \uparrow^{\alpha\beta}$			$\frac{1}{2} k^2 (Y_3 + 4 \binom{(4)}{K_{15}} - 2 \binom{(4)}{K_3})$	0
	$\mathcal{K}_{2^+}^{#1} \uparrow^{\alpha\beta_X}$			0	$\frac{Y_6 k^2}{2}$

	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^+ a\beta\chi}^{\#1}$								
$\mathcal{T}_{0^+}^{\#1} \uparrow$	<table border="1" style="border-collapse: collapse; text-align: center;"> <tr><td style="width: 40px; height: 40px;">1</td><td style="width: 40px; height: 40px;">0</td></tr> <tr><td>Det(0⁺)</td><td></td></tr> </table>	1	0	Det(0 ⁺)		<table border="1" style="border-collapse: collapse; text-align: center;"> <tr><td style="width: 40px; height: 40px;">0</td></tr> </table>	0			
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$\mathcal{T}_{0^+}^{\#1} \uparrow^{a\beta\chi}$	<table border="1" style="border-collapse: collapse; text-align: center;"> <tr><td style="width: 40px; height: 40px;">0</td></tr> </table>	0	<table border="1" style="border-collapse: collapse; text-align: center;"> <tr><td style="width: 40px; height: 40px;">1</td></tr> <tr><td>Det(0⁺)</td></tr> </table>	1	Det(0 ⁺)	$\mathcal{T}_{1^+ a\beta}^{\#1}$	$\mathcal{T}_{1^+ a\beta}^{\#2}$	$\mathcal{T}_{1^+ a}^{\#1}$	$\mathcal{T}_{1^+ a}^{\#2}$	
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	$\mathcal{T}_{1^+}^{\#1} \uparrow^{a\beta}$	<table border="1" style="border-collapse: collapse; text-align: center;"> <tr><td style="width: 60px; height: 40px;">2</td><td style="width: 60px; height: 40px;">$\sqrt{2}$</td></tr> <tr><td>3 Det(1⁺)</td><td>3 Det(1⁺)</td></tr> </table>	2	$\sqrt{2}$	3 Det(1 ⁺)	3 Det(1 ⁺)	<table border="1" style="border-collapse: collapse; text-align: center;"> <tr><td style="width: 60px; height: 40px;">0</td></tr> </table>	0	<table border="1" style="border-collapse: collapse; text-align: center;"> <tr><td style="width: 60px; height: 40px;">0</td></tr> </table>	0								
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$$\begin{aligned}
 & \text{Abbreviations used in matrices} \\
 & Y_1 = 3 \overset{(4)}{\underset{4}{K_1}} - 4 \overset{(4)}{\underset{15}{K}} - 2 \overset{(4)}{\underset{16}{K}} + 3 \overset{(4)}{\underset{2}{K_2}} + 2 \overset{(4)}{\underset{3}{K_3}} + \overset{(4)}{\underset{4}{K_4}} \& Y_2 = \overset{(4)}{\underset{15}{K}} - \overset{(4)}{\underset{16}{K}} \& \\
 & Y_3 = 2 \overset{(4)}{\underset{16}{K}} - \overset{(4)}{\underset{4}{K_4}} \& Y_4 = 2 \overset{(4)}{\underset{15}{K}} + \overset{(4)}{\underset{16}{K}} - 2 \overset{(4)}{\underset{2}{K_2}} \& Y_5 = 2 \overset{(4)}{\underset{2}{K_2}} + \overset{(4)}{\underset{7}{K_7}} \& \\
 & Y_6 = 2 \overset{(4)}{\underset{2}{K_2}} + 2 \overset{(4)}{\underset{3}{K_3}} + \overset{(4)}{\underset{4}{K_4}} + 2 \overset{(4)}{\underset{7}{K_7}} \& Y_7 = 4 \overset{(4)}{\underset{15}{K}} + 2 \overset{(4)}{\underset{16}{K}} - 2 \overset{(4)}{\underset{3}{K_3}} - \overset{(4)}{\underset{4}{K_4}} \& Y_8 = 2 \overset{(4)}{\underset{15}{K}} + \overset{(4)}{\underset{16}{K}} \& \\
 & \text{Det}(0^+) = \frac{1}{2} (-k^2 (3 \overset{(4)}{\underset{15}{K}} - 4 \overset{(4)}{\underset{16}{K}} + 3 \overset{(4)}{\underset{2}{K_2}} + 2 \overset{(4)}{\underset{3}{K_3}} + \overset{(4)}{\underset{4}{K_4}}) - 3 \overset{(2)}{\underset{3}{K_3}}) \& \\
 & \text{Det}(1^+) = \frac{3}{4} k^2 (-2 \overset{(4)}{\underset{16}{K}} + \overset{(4)}{\underset{4}{K_4}}) + 3 \overset{(2)}{\underset{1}{K_1}} \& \text{Det}(2^+) = \frac{1}{2} k^2 (4 \overset{(4)}{\underset{15}{K}} + 2 \overset{(4)}{\underset{16}{K}} - 2 \overset{(4)}{\underset{3}{K_3}} - \overset{(4)}{\underset{4}{K_4}})
 \end{aligned}$$

$$\begin{aligned}
& \text{Lagrangian} \\
& \frac{(2)}{\kappa_1} \mathcal{K}_{a\beta\chi} \mathcal{K}^{a\beta\chi} - 2 \frac{(2)}{\kappa_1} \mathcal{K}^{a\beta\chi} \mathcal{K}_{\beta a\chi} + \frac{(2)}{\kappa_3} \mathcal{K}^{a\beta} \mathcal{K}^{\chi}_{\beta\chi} + \frac{(4)}{\kappa_1} \partial_{\beta} \mathcal{K}^{\delta}_{\chi\delta} \partial^{\chi} \mathcal{K}^{a,\beta} + \frac{(4)}{\kappa_2} \partial_{\chi} \mathcal{K}^{\delta}_{\beta\delta} \partial^{\chi} \mathcal{K}^{a,\beta} + \\
& \frac{(4)}{\kappa_3} \partial_{\beta} \mathcal{K}^{a\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}_{a\chi} + \frac{(4)}{\kappa_4} \partial_{\delta} \mathcal{K}^{a\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\chi} - 2 \frac{(4)}{\kappa_1} \partial_{\beta} \mathcal{K}^{a\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}_{\chi a} - \frac{(4)}{\kappa_1} \partial_{\beta} \mathcal{K}^{a\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}_{\chi a} + \\
& \frac{(4)}{\kappa_3} \partial_{\beta} \mathcal{K}^{a\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}_{\chi a} + \frac{(4)}{\kappa_4} \partial_{\delta} \mathcal{K}^{a\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}_{\chi a} + \frac{(4)}{\kappa_7} \partial^{\chi} \mathcal{K}^{a,\beta} \partial_{\delta} \mathcal{K}^{\delta}_{\chi\beta} - \frac{(4)}{\kappa_1} \partial_{\delta} \mathcal{K}^{a\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\chi} \\
& \frac{1}{2} \frac{(4)}{\kappa_1} \partial_{\delta} \mathcal{K}^{a\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\chi} + \frac{1}{4} \frac{(4)}{\kappa_4} \partial_{\delta} \mathcal{K}^{a\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\chi} + \frac{(4)}{\kappa_1} \partial_{\delta} \mathcal{K}^{a\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\chi} + \frac{(4)}{\kappa_1} \partial_{\delta} \mathcal{K}^{a\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\chi}
\end{aligned}$$

Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{I}_1^{\#1\alpha\beta} =: \mathcal{I}_1^{\#2\alpha\beta}$	3	$3\partial_\alpha\partial_\chi\partial^\alpha\mathcal{I}^{\chi\beta\delta} + \partial_\alpha\partial^\delta\partial_\chi\mathcal{I}^{\beta\alpha\chi} + 2\partial_\alpha\partial^\delta\partial_\chi\mathcal{I}^{\chi\alpha\beta} =: 3\partial_\alpha\partial_\chi\partial^\beta\mathcal{I}^{\chi\alpha\delta} + \partial_\alpha\partial^\delta\partial_\chi\mathcal{I}^{\alpha\beta\chi}$
Total # constraint(s):	3	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{6 \kappa_{15}^{(4)} + 3 \kappa_{16}^{(4)} - 2 \kappa_3^{(4)} - \kappa_4^{(4)}}{(2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) (4 \kappa_{15}^{(4)} + 2 \kappa_{16}^{(4)} - \kappa_3^{(4)} - \kappa_4^{(4)})}$
	2	0	$\frac{-6 \kappa_{15}^{(4)} - 3 \kappa_{16}^{(4)} + 2 \kappa_3^{(4)} + \kappa_4^{(4)}}{(2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) (4 \kappa_{15}^{(4)} + 2 \kappa_{16}^{(4)} - \kappa_3^{(4)} - \kappa_4^{(4)})}$
	2	0	$\frac{-32 \kappa_{15}^{(4)2} - 32 \kappa_{15}^{(4)} \kappa_{16}^{(4)} - 8 \kappa_{16}^{(4)2} + 14 \kappa_{15}^{(4)} (2 \kappa_3^{(4)} + \kappa_4^{(4)}) + 7 \kappa_{16}^{(4)} (2 \kappa_3^{(4)} + \kappa_4^{(4)}) - 2 (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2}{(2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) (2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)} - 2 \kappa_3^{(4)} - \kappa_4^{(4)}) (4 \kappa_{15}^{(4)} + 2 \kappa_{16}^{(4)} - \kappa_3^{(4)} - \kappa_4^{(4)})}$



1

0

$$\begin{aligned}
& -((-(128 \kappa_{15}^4 p^2 - 256 \kappa_{15}^3 \kappa_{16} p^2 - 192 \kappa_{15}^2 \kappa_{16}^2 p^2 - 64 \kappa_{15} \kappa_{16}^3 \kappa_3 p^2 - \\
& 8 \kappa_{16}^4 p^2 - 64 \kappa_{15}^3 \kappa_3 p^2 - 96 \kappa_{15}^2 \kappa_{16} \kappa_3 p^2 - 48 \kappa_{15}^4 \kappa_{16}^2 \kappa_3 p^2 - 8 \kappa_{15}^3 \kappa_{16}^3 \kappa_3 p^2 + \\
& 32 \kappa_{15}^2 \kappa_3^2 p^2 + 32 \kappa_{15}^4 \kappa_{16} \kappa_3^2 p^2 + 8 \kappa_{16}^4 \kappa_3^2 p^2 + 16 \kappa_{15}^4 \kappa_3^3 p^2 + 8 \kappa_{16}^4 \kappa_3^3 p^2 - \\
& 32 \kappa_{15}^3 \kappa_4 p^2 - 48 \kappa_{15}^2 \kappa_{16} \kappa_4 p^2 - 24 \kappa_{15}^4 \kappa_{16} \kappa_4 p^2 - 4 \kappa_{16}^3 \kappa_4 p^2 + 32 \kappa_{15}^2 \kappa_{16}^2 \kappa_4 p^2 + \\
& 32 \kappa_{15}^4 \kappa_{16} \kappa_3 \kappa_4 p^2 + 8 \kappa_{16}^2 \kappa_3 \kappa_4 p^2 + 24 \kappa_{15}^4 \kappa_{16} \kappa_3 \kappa_4 p^2 + 12 \kappa_{16}^4 \kappa_3 \kappa_4 p^2 + 8 \kappa_{15}^2 \kappa_4^2 p^2 + \\
& 8 \kappa_{15}^4 \kappa_{16} \kappa_4^2 p^2 + 2 \kappa_{16}^2 \kappa_4^2 p^2 + 12 \kappa_{15}^4 \kappa_3 \kappa_4^2 p^2 + 6 \kappa_{16}^4 \kappa_3 \kappa_4^2 p^2 + 2 \kappa_{15}^2 \kappa_4^3 p^2 + \\
& \kappa_{16}^2 \kappa_4^3 p^2 - 192 \kappa_{15}^3 \kappa_7 p^2 - 288 \kappa_{15}^2 \kappa_{16} \kappa_7 p^2 - 144 \kappa_{15}^4 \kappa_{16}^2 \kappa_7 p^2 - 24 \kappa_{16}^4 \kappa_7 p^2 + \\
& 48 \kappa_{15}^4 \kappa_3 \kappa_7 p^2 + 24 \kappa_{16}^4 \kappa_3 \kappa_7 p^2 + 48 \kappa_{15}^3 \kappa_3 \kappa_4 \kappa_7 p^2 + 24 \kappa_{16}^3 \kappa_3 \kappa_4 \kappa_7 p^2 + \\
& 12 \kappa_{15}^4 \kappa_4 \kappa_7 p^2 + 6 \kappa_{16}^4 \kappa_4 \kappa_7 p^2 - 72 \kappa_{15}^2 \kappa_7 p^2 - 72 \kappa_{15}^4 \kappa_{16} \kappa_7 p^2 - 18 \kappa_{16}^2 \kappa_7 p^2 + \\
& 36 \kappa_{15}^4 \kappa_3 \kappa_7 p^2 + 18 \kappa_{16}^4 \kappa_3 \kappa_7 p^2 + 18 \kappa_{15}^3 \kappa_4 \kappa_7 p^2 + 9 \kappa_{16}^3 \kappa_4 \kappa_7 p^2 - \\
& \sqrt{3} \sqrt{((2 \kappa_{15}^4 + \kappa_{16}^2)^2 (4096 \kappa_{15}^6 p^4 + 64 \kappa_{16}^6 p^4 - 64 \kappa_{16}^5 (2 \kappa_3 + \kappa_4 - 4 \kappa_7) p^4 + 2048 \kappa_{15}^5 (6 \kappa_{16} - \\
& 2 \kappa_3 - \kappa_4 + 4 \kappa_7) p^4 + \kappa_{16}^4 (147 \kappa_3^2 + 16 (4 \kappa_3^2 + \kappa_4^2 + 4 \kappa_3 (\kappa_4 - 8 \kappa_7) - 16 \kappa_4 \kappa_7 + 24 \kappa_7^2) p^4) + \\
& 16 \kappa_{15}^3 (294 \kappa_{16} \kappa_3^2 - 462 \kappa_3 \kappa_3^2 - 231 \kappa_4 \kappa_3^2 + 640 \kappa_{16}^3 p^4 + 128 \kappa_3^3 p^4 + 16 \kappa_{16}^4 p^4 - 320 \kappa_{16}^2 \\
& (2 \kappa_3 + \kappa_4 - 4 \kappa_7) p^4 + 80 \kappa_4^2 \kappa_7 p^4 - 144 \kappa_4 \kappa_7^2 p^4 + 144 \kappa_7^3 p^4 + 64 \kappa_3^2 (3 \kappa_4 + 5 \kappa_7) p^4 + 32 \kappa_3^3 \\
& (3 \kappa_4 + 10 \kappa_4 \kappa_7 - 9 \kappa_7^2) p^4 + 32 \kappa_{16}^4 (4 \kappa_3^2 + \kappa_4^2 + 4 \kappa_3 (\kappa_4 - 8 \kappa_7) - 16 \kappa_4 \kappa_7 + 24 \kappa_7^2) p^4) + \\
& 16 \kappa_{15}^4 (147 \kappa_3^2 + 16 (60 \kappa_{16}^2 + 4 \kappa_3^2 + \kappa_4^2 + 4 \kappa_3 (\kappa_4 - 8 \kappa_7) - 20 \kappa_{16} (2 \kappa_3 + \kappa_4 - 4 \kappa_7) - 16 \\
& \kappa_4 \kappa_7 + 24 \kappa_7^2) p^4) + 2 \kappa_{16}^3 (-462 \kappa_3^3 p^4 + 128 \kappa_3^3 p^4 + 16 \kappa_4^3 p^4 + 80 \kappa_4^2 \kappa_7 p^4 + 144 \kappa_7^3 p^4 + \\
& 64 \kappa_3^2 (3 \kappa_4 + 5 \kappa_7) p^4 + 32 \kappa_3^2 (3 \kappa_4 + 10 \kappa_4 \kappa_7 - 9 \kappa_7^2) p^4 - 3 \kappa_4 (77 \kappa_3^2 + 48 \kappa_7^2) p^4) - \\
& 4 \kappa_{16}^2 (2 \kappa_3 + \kappa_4) (16 \kappa_3^4 p^4 + \kappa_4^4 p^4 + 14 \kappa_4^3 \kappa_7 p^4 + 18 \kappa_4^2 \kappa_7^2 p^4 + 27 \kappa_4 \kappa_7^3 p^4 + \\
& 16 \kappa_3^3 (2 \kappa_4 + 7 \kappa_7) p^4 + 6 \kappa_4 (11 \kappa_3^2 + 6 \kappa_7^2 p^4) + 4 \kappa_3 (66 \kappa_4 \kappa_3 + 2 \kappa_4^3 p^4 + 21 \kappa_4^2 \kappa_7 p^4 + \\
& 36 \kappa_4 \kappa_7^2 p^4 + 9 \kappa_7^3 p^4) + 24 \kappa_3^2 (11 \kappa_3^2 + (\kappa_4 + 7 \kappa_4 \kappa_7 + 6 \kappa_7^2) p^4) + (2 \kappa_3 + \kappa_4)^2 \\
& (48 \kappa_3^4 p^4 + 3 \kappa_4^4 p^4 + 20 \kappa_4^3 \kappa_7 p^4 + 36 \kappa_4^2 \kappa_7^2 p^4 + 27 \kappa_4 \kappa_7^3 p^4 + 32 \kappa_3^3 (3 \kappa_4 + 5 \kappa_7) p^4 + \\
& 6 \kappa_4^2 (8 \kappa_3^2 + 7 \kappa_7^2 p^4) + 24 \kappa_3 (8 \kappa_4 \kappa_3 + \kappa_4^3 p^4 + 5 \kappa_4^2 \kappa_7 p^4 + 7 \kappa_4 \kappa_7^2 p^4 + 3 \kappa_7^3 p^4) + \\
& 24 \kappa_3^2 (8 \kappa_3^2 + (3 \kappa_4 + 10 \kappa_4 \kappa_7 + 7 \kappa_7^2) p^4) + 4 \kappa_{15}^2 (531 \kappa_4 \kappa_3 + 960 \kappa_{16} p^4 - 320 \kappa_3^4 p^4 - \\
& 20 \kappa_4^4 p^4 - 640 \kappa_{16}^3 (2 \kappa_3 + \kappa_4 - 4 \kappa_7) p^4 - 16 \kappa_4^3 \kappa_7 p^4 + 168 \kappa_4 \kappa_7^2 p^4 - 144 \kappa_4 \kappa_3^2 p^4 + \\
& 108 \kappa_7^2 p^4 - 128 \kappa_3^3 (5 \kappa_4 + \kappa_7) p^4 + 12 \kappa_3^2 (177 \kappa_3^2 - 8 (5 \kappa_4 + 2 \kappa_4 \kappa_7 - 7 \kappa_7^2) p^4) + \\
& 6 \kappa_{16}^2 (147 \kappa_3^2 + 16 (4 \kappa_3^2 + \kappa_4^2 + 4 \kappa_3 (\kappa_4 - 8 \kappa_7) - 16 \kappa_4 \kappa_7 + 24 \kappa_7^2) p^4) + \\
& 6 \kappa_{16}^3 (-462 \kappa_3 \kappa_3^2 + 128 \kappa_3^3 p^4 + 16 \kappa_4^3 p^4 + 80 \kappa_4^2 \kappa_7 p^4 + 144 \kappa_7^3 p^4 + \\
& 64 \kappa_3^2 (3 \kappa_4 + 5 \kappa_7) p^4 + 32 \kappa_3^2 (3 \kappa_4 + 10 \kappa_4 \kappa_7 - 9 \kappa_7^2) p^4 - 3 \kappa_4 (77 \kappa_3^2 + 48 \kappa_7^2) p^4) - \\
& 4 \kappa_3^2 (40 \kappa_4^3 p^4 + 24 \kappa_4^2 \kappa_7 p^4 + 72 \kappa_4 \kappa_7^2 p^4 - 3 \kappa_4 (177 \kappa_3^2 + 56 \kappa_7^2) p^4) - \\
& \kappa_{16}^2 (320 \kappa_3^3 p^4 + 20 \kappa_4^4 p^4 + 16 \kappa_4^3 \kappa_7 p^4 + 144 \kappa_4^2 \kappa_7^2 p^4 - 108 \kappa_4 \kappa_7^3 p^4 + 128 \kappa_3^2 (5 \kappa_4 + \kappa_7) p^4 - \\
& 3 \kappa_4^2 (177 \kappa_3^2 + 56 \kappa_7^2) p^4 - 12 \kappa_3^2 (177 \kappa_3^2 - 8 (5 \kappa_4 + 2 \kappa_4 \kappa_7 - 7 \kappa_7^2) p^4) + \\
& 4 \kappa_3^3 (40 \kappa_4^3 p^4 + 24 \kappa_4^2 \kappa_7 p^4 + 72 \kappa_4 \kappa_7^2 p^4 - 3 \kappa_4 (177 \kappa_3^2 + 56 \kappa_7^2) p^4) + \\
& 4 \kappa_{15}^2 (192 \kappa_{16}^5 p^4 - 160 \kappa_{16}^4 (2 \kappa_3 + \kappa_4 - 4 \kappa_7) p^4 +$$

[illegible]

	1	$-\frac{3 \binom{2}{k_3}}{3 \kappa_1 - 4 \binom{(4)}{k_1} \kappa_{15} - 2 \binom{(4)}{k_{16}} + 3 \binom{(4)}{k_2} + 2 \binom{(4)}{k_3} + \binom{(4)}{k_4}}$	$-\frac{2}{3 \kappa_1 - 4 \binom{(4)}{k_1} \kappa_{15} - 2 \binom{(4)}{k_{16}} + 3 \binom{(4)}{k_2} + 2 \binom{(4)}{k_3} + \binom{(4)}{k_4}}$
	1	$\frac{3 \binom{2}{k_3}}{-\binom{(4)}{k_{15}} + \kappa_{16}}$	$\frac{1}{\binom{(4)}{k_{15}} + \kappa_{16}}$
	3	$\frac{4 \binom{2}{k_3}}{2 \binom{(4)}{k_{15}} - \binom{(4)}{k_4}}$	$-\frac{4}{6 \binom{(4)}{k_{16}} - 3 \binom{(4)}{k_4}}$
	3	$2(-2 \binom{(4)}{k_{15}} \binom{(4)}{k_{16}} + 2 \binom{(4)}{k_3} \binom{(4)}{k_4} + \binom{(2)}{k_3})$	$\frac{2(8 \binom{(4)}{k_{15}}^2 + 2 \binom{(4)}{k_{16}}^2 + 4 \binom{(4)}{k_3}^2 + 4 \binom{(4)}{k_3} \binom{(4)}{k_4} + 4 \binom{(4)}{k_3} \binom{(4)}{k_{16}} \binom{(4)}{k_7} + 4 \binom{(4)}{k_3} \binom{(4)}{k_7} + 2 \binom{(4)}{k_4} \binom{(4)}{k_7} + 3 \binom{(4)}{k_7} + 8 \binom{(4)}{k_{15}} (\binom{(4)}{k_{16}} + \binom{(4)}{k_7}))}{}$